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PROJECT PORTFOLIO

Selected Client-Context, Academic and Independent Solutions

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DIGITAL TRANSFORMATION, INFORMATION-SECURITY GOVERNANCE AND OPERATIONAL RESILIENCE

Independent ISO/IEC 27001 ISMS Reference Portfolio

August 2026-present - Independent information-security governance reference project

- * **Context:** Needed a structured way to translate ongoing ISO/IEC 27001:2022 Lead Implementer study and previous business and technology analysis experience into inspectable governance work relevant to small, resource-constrained organisations.
- * **Objective:** Develop and progressively publish a portfolio demonstrating how proportionate ISMS foundations could be designed for small organisations operating in different technological and information-risk environments.
- * **Contribution:** Defined the initial purpose and three-scenario structure of the portfolio and began developing the reference framework and supporting materials.
- * **Reference scenarios:** Structured the project around a small VSAT company, a small robotics company and a generic GP practice. These are model organisations rather than clients or representations of commissioned implementations.
- * **Planned ISMS coverage:** Organisational context, interested parties, ISMS boundaries, information-security risks, risk-assessment methodology, control selection, governance documentation and proportionate implementation priorities.
- * **Publication approach:** Intends to publish the cases progressively through a personal professional website, clearly distinguishing completed material, work in progress and planned components.
- * **BIS pathway:** Joined the British Interplanetary Society and initiated contact concerning a possible proportionate information-security governance and ISMS-foundations initiative. An initial conversation took place, but no substantive follow-up, project scope, responsibilities or formal assignment have yet been agreed.
- * **CIIISec pathway:** Joined the Chartered Institute of Information Security and intends to seek participation in an appropriate working group or associated project once the forthcoming groups and participation arrangements are published. No working-group appointment or project assignment has yet been confirmed.
- * **ARPAS-UK pathway:** Membership is scheduled to commence on 5 September 2026 as a prospective route to demonstrable work relevant to the drone sector. Based on information received, membership is expected to provide an opportunity to propose journal contributions. Intends to develop a relevant article and explore whether a proportionate information-security governance initiative could support smaller drone-sector organisations. No article has yet been submitted or accepted, and no ARPAS-UK project has been proposed, commissioned or agreed.
- * **Planned output:** A collection of published reference cases and reusable governance artefacts demonstrating the application of ISO/IEC 27001 principles across satellite communications, robotics and primary-care settings.
- * **Value and status:** The independent portfolio project is active and work has commenced. The BIS opportunity remains exploratory, CIIISec project participation is pending publication of the new working groups, and ARPAS-UK membership is scheduled, while the proposed publication and organisational

initiative remain planned. No external association is presented as having commissioned, approved or validated the independent portfolio.

Open Source, Open Standards and Digital Sovereignty Operating Model Proposal

2024 - Client-context initiative developed during the Link Communications Systems engagement

* **Context:** Identified an opportunity to examine whether a small satellite-connectivity business could reduce supplier dependency, strengthen data control and improve long-term operational ownership through greater use of open source software (OSS), open standards and UK- or EU-hosted services.

* **Objective:** Develop and assess an alternative operating model covering collaboration, file management, CRM, infrastructure and AI services while balancing cost, security, privacy, regulatory considerations, support requirements and organisational capacity.

* **Contribution:** Designed an OSS-first operating model proposal combining managed Nextcloud, UK- or EU-hosted infrastructure and services, open source CRM, and private or EU-hosted AI options.

* **Option assessment:** Compared alternative service and supplier arrangements across functionality, hosting jurisdiction, implementation requirements, support availability, interoperability, data portability, supplier dependency and internal ownership.

* **Software assessment approach:** Drew on the Navica Open Source Maturity Model (N-OSMM) and the broader OSS quality-assessment dimensions described by Spinellis et al. (2009), considering functionality, longevity, product quality, development and maintenance activity, documentation, community and paid support, training, integration and professional-services availability.

* **Financial scenario:** Built a three-year lifecycle scenario identifying approximately £5,000-£28,000 in potential gross savings, while distinguishing these possible savings from implementation, migration, support, training and operational-management costs.

* **Governance assessment:** Examined supplier governance, data control, international data transfers, interoperability, implementation ownership and the organisation's capacity to operate or oversee the proposed environment.

* **Regulatory and security assessment:** Connected the proposal with UK GDPR, international-transfer and PECR considerations, NCSC Cloud Security Principles, EU Data Act principles and potentially relevant NIS2 and UK cyber-resilience considerations for satellite-sector supply chains.

* **Output:** Produced a structured transformation proposal and decision framework intended to enable management to compare OSS, managed-service, conventional cloud and supplier-dependent options across financial, operational, governance and regulatory dimensions.

* **Value and status:** Created a quantified strategic option for strengthening digital sovereignty, resilience and operational control within a resource-constrained satellite-connectivity SME. The estimated savings were modelled rather than realised, and the proposal was not implemented yet.

Open Source, Open Standards and Digital Resilience Roadmap

2024 - Independent follow-on project informed by the North Kesteven District Council engagement

* **Context:** The council engagement identified software duplication, legacy dependencies, specialist operational requirements, differing levels of staff readiness and potential opportunities for greater use of open source software and open standards.

- * **Objective:** Explore how a local-government organisation could progressively reduce avoidable supplier dependency, improve interoperability and data portability, and strengthen long-term digital resilience without assuming that full systems replacement would be operationally or organisationally realistic.
- * **Contribution:** Developed a five-phase open source (OSS), Open standards and Digital resilience roadmap, extending lessons from the original ICT-refresh and software-rationalisation discovery into an independently developed transformation scenario.
- * **Software assessment:** Identified potential Linux-compatible OSS options and adapted the Navica Open Source Maturity Model to assess their maturity and readiness across functionality, longevity, product quality, community and paid support, documentation, training, integration and professional services.
- * **Analysis:** Used weighted maturity scoring, descriptive statistics, chi-square testing and hierarchical clustering to examine patterns, strengths and red flags across the assessed OSS options and inform the transformation roadmap.
- * **Output:** Produced a retrospective three-year scenario exploring approximately £35,000-£240,000 in possible gross avoided-cost exposure while considering the practical requirements and constraints associated with implementation.
- * **Governance assessment:** Examined vendor dependency, interoperability, data portability, implementation ownership and organisational readiness as factors affecting whether potential technical and financial benefits could be realised.
- * **Value and status:** Created a structured strategic option for examining longer-term digital resilience and supplier-dependency reduction in a local-government context. The project was independently developed after the engagement, was not commissioned or implemented by the council, and the estimated value represents possible gross avoided-cost exposure rather than realised savings.

Digital Transformation Journey Assessment for British Councils

2024 - Independent public-sector research and thought-leadership project informed by the North Kesteven District Council engagement

- * **Context:** Used insights from a district-council ICT engagement to examine wider organisational conditions affecting digital transformation across British local authorities.
- * **Objective:** Assess how councils could balance institutional stability, specialist public-service requirements, political accountability and financial stewardship with the need for greater integration, adaptability and digital innovation.
- * **Contribution:** Developed a comparative current-state and improvement-opportunity assessment covering seven dimensions of council culture, structure, governance and digital transformation readiness.
- * **Assessment:** Examined stability and risk aversion, departmental specialisation, cross-functional integration, organisational agility, political considerations, allocation of public resources and the protective responsibilities associated with public-service delivery.
- * **Transformation approach:** Proposed a human-centred model combining multidisciplinary capability, organisational culture, digital awareness, adaptability, ethics, sustainability, security, privacy and modular technology choices.

* **Output:** Authored and published a structured case-based article translating experience from one district-council engagement into a broader assessment framework for British local-government digital transformation.

* **Value and status:** Demonstrated the ability to convert operational experience into transferable public-sector analysis and strategic thought leadership. The work was independently developed and published; it was not commissioned, validated or implemented by a council.

Network Operations Centre Improvement Framework

2023 - Client-context initiative developed during the Link Communications Systems engagement

* **Context:** A four-person, experienced Network Operations Centre team presented potential concerns involving organisational alignment, customer care, procedures and change readiness, while management wished to minimise the operational risks associated with changes to established personnel arrangements.

* **Objective:** Develop a structured route for assessing the existing NOC and identifying proportionate improvements across people, processes, technology and service delivery.

* **Contribution:** Designed a multi-phase framework covering current-state assessment, target setting, change identification, budgeting, implementation and service transition.

* **Output:** Structured the assessment around organisational culture, customer care, personnel, procedures, software, hardware, reliability, service standards, training, supplier selection and implementation requirements.

* **Value and status:** Created a preliminary transformation framework that recognised management risk tolerance, operational continuity and the need for direct NOC familiarisation before final recommendations. The framework was developed as a proposal and was not implemented within the engagement.

SPACE-SECTOR INTELLIGENCE AND RESEARCH AUTOMATION

European Space Agency-Linked Space Industry Discovery AI Agent

2026 - Independent AI-assisted research-automation project

* **Context:** Needed a scalable method for identifying possible commercial organisations appearing across fragmented official European and national space-sector sources, while preserving sufficient evidence for later verification and targeted outreach.

* **Objective:** Develop a local, open source (OSS) and repeat-safe discovery AI-assisted automation pipeline capable of collecting candidate organisation names at high recall without presenting unverified names as confirmed companies, national entities and contractors composing the UK and Italian ecosystem of the European Space Agency (ESA).

* **Role and contribution:** Defined the research requirements, source scope, workflow stages, validation rules, data structures and quality controls; directed and validated the ChatGPT-assisted development of the Python and SQLite implementation. Did not personally author the code.

* **Sources:** Structured the workflow around official ESA, UK Space Agency, Italian Space Agency, ESA Business Incubation Centre, ESA commercialisation and Italian Space Industry catalogue sources.

- * **Workflow design:** Combined controlled web crawling, HTML and readable-text extraction, sitemap and WordPress REST discovery, local Ollama-assisted name extraction, deterministic validation and SQLite-based persistence.
- * **Data-governance approach:** Required preservation of source lineage, supporting evidence and processing histories, while preventing unsupported claims concerning company location, commercial status, contractor status or the nature of an organisation's relationship with ESA.
- * **Reliability controls:** Incorporated database backups, restart and interruption recovery, repeat-safe processing, targeted regression checks, permitted-domain controls, audit histories and resumable batch execution.
- * **Data structure:** Specified structured records for source pages, candidate names, aliases, organisation relationships, validation decisions and supporting evidence.
- * **Output:** Produced high-recall datasets containing 1,663 raw UK name candidates and 372 Italian name candidates, including 302 profiles harvested from the Italian Space Industry catalogue.
- * **Value and status:** Created a reusable research pipeline and evidence base for subsequent company verification, market research and targeted outreach. The project intentionally prioritised recall, and the final determination of which candidates were genuine commercial companies remained a manual filtering task that was not completed within the automated workflow.

UK and EU Rural-Connectivity Opportunity Analysis

2023 - Client-context initiative developed during the Link Communications Systems engagement

- * **Context:** Identified an opportunity to examine whether public connectivity data could support a more evidence-based approach to rural satellite and VSAT business development.
- * **Objective:** Assess connectivity gaps, underserved user groups and locations where terrestrial coverage limitations could create plausible demand for satellite-connectivity services.
- * **Contribution:** Gathered and compared public UK and EU coverage information, examined potential customer groups and translated the findings into a commercial opportunity-assessment framework.
- * **Output:** Developed a data-led proof of concept connecting geographic coverage gaps with potential rural-connectivity, organisational and public-service use cases.
- * **Value and status:** Created a reusable basis for market prioritisation and further feasibility analysis. The project was at proof-of-concept stage and did not progress to commercial implementation during the engagement.

DATA, AI AND SERVICE-DESIGN CONCEPTS

VogueVictory: Privacy-Aware Fashion Gamification Concept

2024 - Independent product and business-model concept | RedDog Transformation

- * **Context:** Explored how fashion brands could use gamification to create forms of customer engagement extending beyond conventional browsing, purchasing and marketing interactions.

- * **Objective:** Conceptualise a platform that could strengthen participation, community interaction and brand loyalty while generating business-intelligence insights into customer preferences and emerging fashion trends.
- * **Solution concept:** Designed VogueVictory as a dedicated gamification area integrated into an existing fashion brand's website or application, where users could respond to themed creative prompts, participate in interactive activities, share personal expression and receive recognition or rewards.
- * **User proposition:** Envisaged an interactive environment in which fashion enthusiasts could engage with brands and other participants, contribute creative content and receive community feedback and recognition.
- * **Business proposition:** Proposed using participation, interaction and voting data to help fashion brands identify customer preferences, understand emerging trends and inform future marketing, engagement and collection strategies.
- * **Privacy approach:** Incorporated anonymous interaction-data collection as a core principle, intended to protect personal information while allowing aggregated participation data to support business-intelligence analysis.
- * **Service model:** Combined the proposed digital platform with an associated consultancy service intended to help brands interpret participation data, develop tailored engagement strategies and align the concept with their business objectives.
- * **Anticipated value:** Identified potential benefits including stronger customer engagement, creativity, community formation, brand loyalty and data-informed strategic decision-making.
- * **Value and status:** Produced and published an initial privacy-aware product and business-model concept connecting gamification, fashion engagement and business intelligence. The concept was not developed into a technical architecture or prototype and was not tested, commercially validated or implemented.

AI-Assisted GP Appointment Prioritisation

2022 - individual academic digital transformation project | University of Hull

- * **Context:** Investigated prolonged GP appointment waiting times in England, rising patient demand, workforce pressure, inconsistent prioritisation practices and the risk that delays could contribute to worsening health conditions, missed appointments and diagnostic errors.
- * **Objective:** Explore whether a customer-led digital platform and machine-learning-supported prioritisation process could reduce waiting times and workload while preserving clinical oversight and access for patients unable or unwilling to use digital channels.
- * **Contribution:** Researched the causes, operational constraints and risks associated with GP waiting times, drawing on NHS guidance, healthcare research, workforce evidence, patient-access data and literature on machine learning, teamwork and technology adoption.
- * **Solution design:** Developed a conceptual platform through which patients could submit requests by app, telephone or direct contact and receive an initial prioritisation based on urgency and the most suitable consultation channel: text, telephone or face-to-face appointment.
- * **Process design:** Produced a cross-functional flowchart mapping interactions between patients, nurse receptionists, doctors, the Royal College of General Practitioners and the machine-learning component, while retaining human review and clinical authority where required.

- * **Data and machine-learning concept:** Proposed the gradual use of machine learning to analyse request information, support urgency classification and learn from subsequent decisions and outcomes, using reinforcement-learning and natural-language-understanding concepts.
- * **Organisational assessment:** Examined the importance of role clarity, trust, communication, teamwork, staff training and inclusive implementation in supporting adoption and reducing resistance or additional pressure on GP personnel.
- * **Inclusion and ethics:** Assessed barriers affecting older patients, including limited smartphone familiarity, visual constraints and affordability, and retained telephone and direct staff-assisted routes to prevent digital exclusion.
- * **Risk assessment:** Considered false positives, model degradation, inappropriate reliance on automated decisions, fragmented technology standards and differences between NHS regions, concluding that machine learning should support rather than replace accountable human judgement.
- * **Output:** Produced a 19-page academic report, conceptual architecture, stakeholder-process flowchart and evidence-based assessment of the operational, technological, organisational and ethical implications of the proposed solution.
- * **Value and status:** Demonstrated the translation of a public-service problem into a structured Digital transformation and service-design concept integrating process redesign, stakeholder collaboration, data analytics and responsible AI. The proposal remained an academic concept and was not clinically tested or implemented.

INNOVATION AND CONTINUOUS-IMPROVEMENT RESEARCH

Lean Six Sigma and Innovation Integration Research

2021 - Undergraduate major research project | Anglia Ruskin University

- * **Context:** Investigated the relationship between Lean Six Sigma, continuous improvement and different forms of innovation, addressing the absence of a single agreed framework explaining where the approaches intersect, support one another or create tensions.
- * **Objective:** Develop a practical reference that could help innovation and Lean Six Sigma practitioners identify the most appropriate improvement approaches, frameworks and tools for different innovation contexts.
- * **Method:** Conducted a qualitative, secondary-literature investigation using an inductive and comparative approach, reviewing freely accessible academic and professional sources concerning innovation, Lean Six Sigma, continuous improvement, design, quality management and organisational transformation.
- * **Research structure:** Classified 10 innovation types, including process, product, incremental, radical, problem-driven, sustaining, disruptive, breakthrough and architectural innovation, alongside the broader category of innovation capability.
- * **Analytical contribution:** Identified 13 areas of commonality between innovation and Lean Six Sigma, covering customer needs, lifecycle, motivation, design, organisational weaknesses, decision rights, frameworks, culture, complexity, scope, supply chains, final impact and enhancement.

*** Framework design:** Cross-referenced the innovation types and shared areas against relevant Lean Six Sigma applications, including DMAIC, Design for Six Sigma, Value Stream Mapping, Quality Function Deployment, TRIZ, Lean risk management, Lean CRM, Design Thinking and business-intelligence approaches.

*** Output:** Produced a 54-page research report, an innovation-type and Lean Six Sigma integration matrix, a comparative analysis table and a bar chart showing the number of identified Lean Six Sigma applications associated with each innovation type.

*** Findings:** Found that innovation capability and product innovation had the highest level of identified interaction with Lean Six Sigma, while architectural innovation showed the lowest. Process, incremental, radical, problem-driven, sustaining and disruptive innovation occupied a broadly similar middle range.

*** Value and status:** Created an original decision-support reference for selecting improvement tools according to the innovation context. The work was an academic literature-based framework and was not tested through organisational implementation.